AiSentinel Dashboard

AI-Powered Cybersecurity Platform

Course: Mobile Computing

Supervisors: Dr. Ahmed Hussien / Eng.Rania El-Sayed

Team Member	UX Research&Wireframe	Frontend	Backend	Report
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Hager Essam Gaber Mahmoud	✔	✔		✔
Rama Ehab Ahmed Said	✔	✔		✔
Seif Ashraf Fawzy Roushdy	✔	✔		✔

Project Introduction

- **Concept:** An AI-powered cybersecurity platform designed to detect, analyze, and respond to threats in real time. The system combines rule-based detection with machine learning models to identify known attacks, detect anomalies, and automate incident response through a unified SOC dashboard.

- **Problem Statement:** Traditional monitoring tools often lack automated security for administrative access, leading to unauthorized entry. Additionally, static log analysis is slow and fails to keep up with real-time threats.
- **Proposed Solutions:**
  - **RBAC Implementation:** Role-Based Access Control to separate Admin and Analyst duties.
  - **Security Automation:** Automated password updates for new users to ensure immediate initial security.
  - **AI-Driven Real-time Dashboard:** Instant categorization of logs into 'Normal' or 'Attack' using machine learning models.

Section 01: UX Phase (Subsections: UX stage I, II, III, and IV)

a. Stage I: Research & Survey Analysis

Research Method: Online Survey via Google Forms

Sample Size: 8 Participants

Target: SOC Analysts, Security Engineers, IT/CS Students

Key Survey Questions & Findings:

Question	Top Answer	Percentage
How often do you experience alert fatigue?	Sometimes to Often	~75%
Biggest frustration with current tools?	False positives / Lack of automation	~62%
Most important dashboard feature?	Real-time alerts & Automation	~80%
AI features most desired?	Auto threat detection & Risk Scoring	~70%
Trust in AI for security decisions (1–5)?	3.5 average	Moderate trust
Preferred UI mode?	Dark Mode + Customizable layout	~85%

Pain Points: Alert fatigue, difficulty distinguishing real threats from false positives, lack of automation.

Gains: Demand for real-time alerts, AI-driven detection, IP auto-blocking, and risk scoring.

Screenshots

AiSentinel

Section 1 of 3

AiSentinel Dashboard - User Survey

We are building AiSentinel, an AI-powered cybersecurity platform designed to detect, analyze, and respond to threats in real time. The system combines rule-based detection with machine learning models to identify known attacks, detect anomalies, and automate incident response through a unified SOC dashboard.
This survey aims to gather insights from security and tech professionals to design a more intuitive, useful, and intelligent monitoring experience. Your feedback will directly shape the dashboard features, usability, and overall user experience.

This form is automatically collecting emails from all respondents. Change settings

Full Name (optional)

Short answer text

What is your current role? *

- ○ Cybersecurity / Engineering Student
- ○ Security Operations (SOC / Incident Response)
- ○ Security Engineer
- ○ IT / System Administrator
- ○ Security Manager / Team Lead
- ○ Other: ________

After section 1 Continue to next section

Section 2 of 3

Section title (optional)

Description (optional)

Have you ever used security monitoring systems before? *

- ◯ Yes, extensively
- ◯ Yes, but only occasionally or in a limited way
- ◯ No, I haven't

What tools do you currently use for monitoring security? *

- ◯ Splunk
- ◯ IBM QRadar
- ◯ Custom/In-house Dashboard
- ◯ Other: ____________

How often do you face alert fatigue? *

- Rare
- Sometimes
- Often
- Always

What is the most frustrating part of current security dashboards? *

- Cluttered interface / Too much data
- Lack of automation in response
- Difficulty in distinguishing real threats from false positives
- Slow performance and lagging
- Other:

After section 2 Continue to next section

Section 3 of 3

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Description (optional)

What matters most in a security dashboard? *

- Real-time alerts
- Visual graphs
- Simplicity
- Automation
- Customization

Which features would you like in an AI-powered SOC dashboard? *

- [ ] Auto threat detection
- [ ] IP auto-blocking
- [ ] Attack timeline
- [ ] Smart recommendations
- [ ] Risk scoring

How do you prefer receiving alerts? *

- ○ In-Dashboard notifications only
- ○ Email notifications
- ○ Instant messaging (Slack / Microsoft Teams)
- ○ Telegram Bot alerts

Should the system auto-block attackers? *

- ○ Yes always
- ○ Only high-risk
- ○ No (manual only)

How much do you trust AI in making security decisions? *

	1	2	3	4	5	
Not at all	○	○	○	○	○	Fully trust

What type of dashboard layout do you prefer? *

- ◯ Minimal
- ◯ Detailed
- ◯ Customizable

Do you prefer dark mode? *

- ◯ Yes
- ◯ No
- ◯ Optional

Any other thoughts or feedback to help us improve the AiSentinel Dashboard ?

Long answer text

Who has responded?

Email
hager20121999@gmail.com
hagar.khaled485@gmail.com
roaa71829@gmail.com
noor.alhoda.eisa@gmail.com
hassanadel0115@gmail.com
mariam.ashraf.eng@gmail.com
hageresam513@gmail.com
d.shahdahmed020@gmail.com

Full Name (optional)

4 responses

shahd

رؤى محمود عبد الوهاب

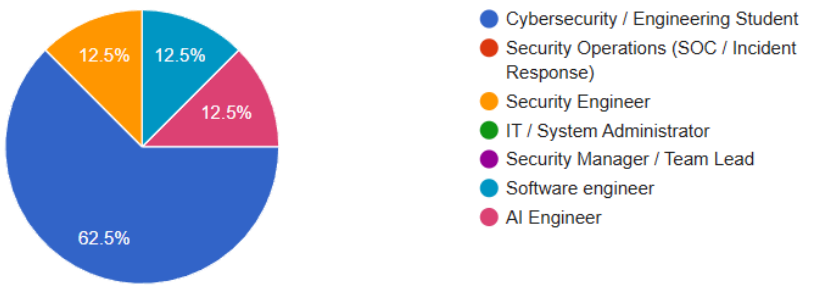
Hagar

HAGAR

What is your current role?

Copy chart

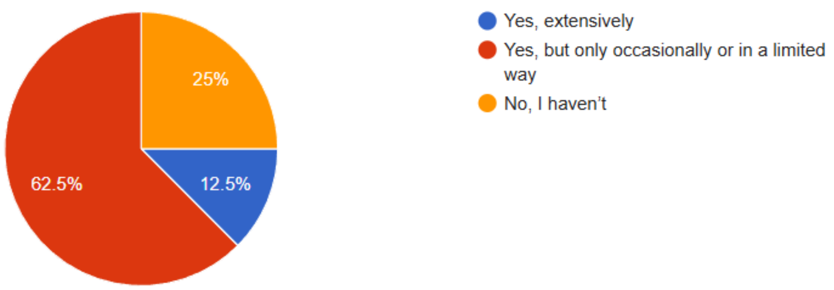
8 responses



Have you ever used security monitoring systems before?

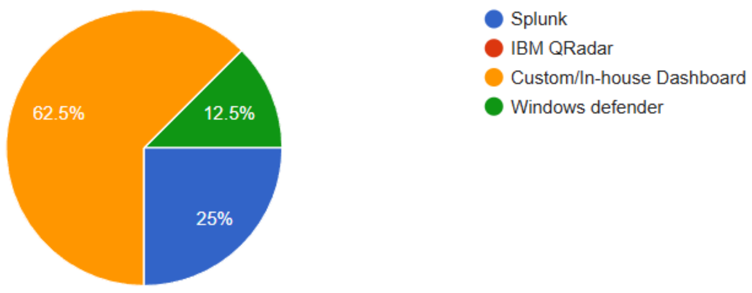
Copy chart

8 responses



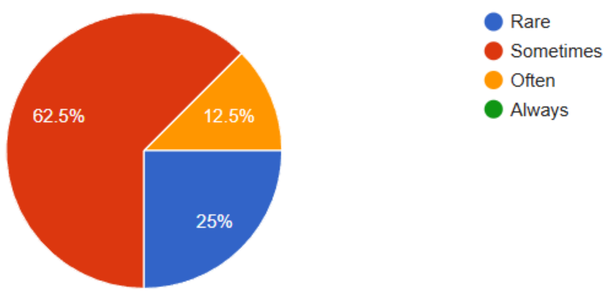
What tools do you currently use for monitoring security?

8 responses



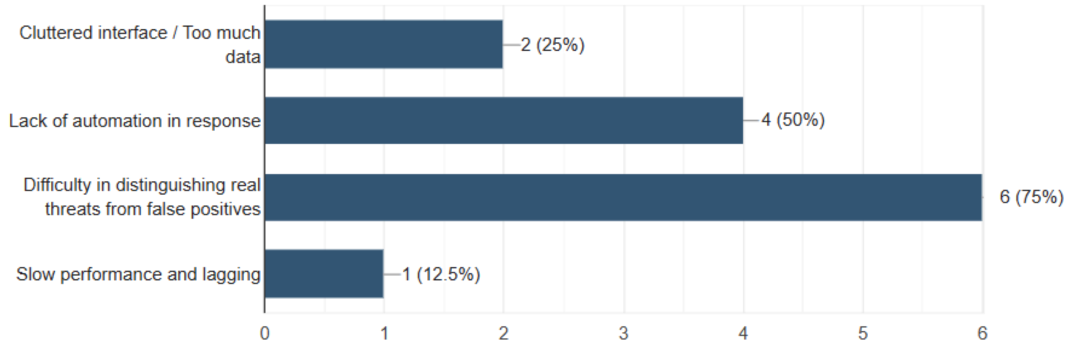
How often do you face alert fatigue?

8 responses



What is the most frustrating part of current security dashboards?

8 responses



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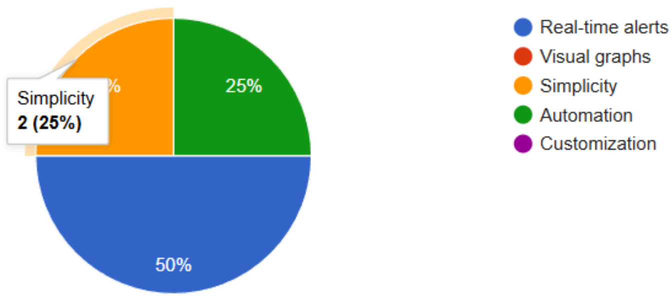
What matters most in a security dashboard?

Copy chart

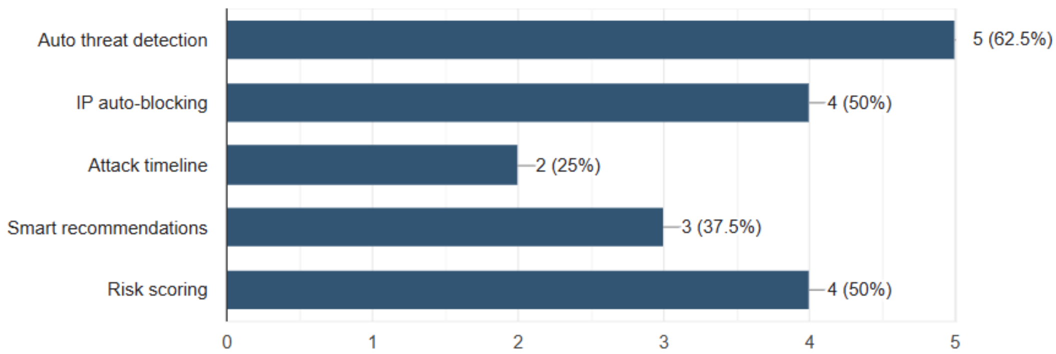
8 responses



Which features would you like in an AI-powered SOC dashboard?

Copy chart

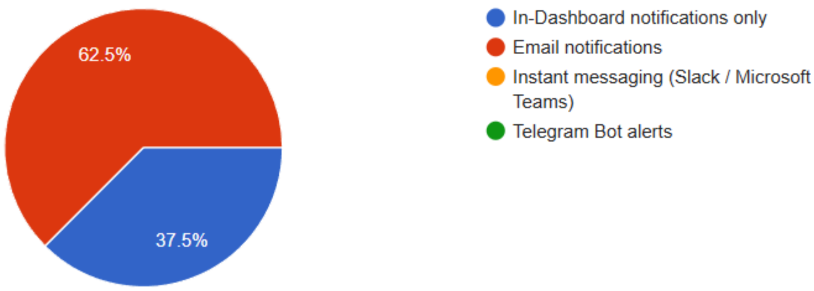
8 responses



How do you prefer receiving alerts?

Copy chart

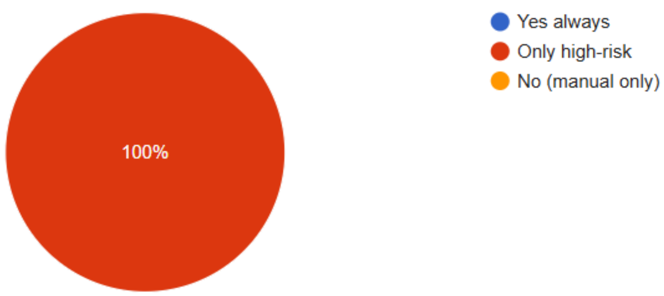
8 responses



Should the system auto-block attackers?

Copy chart

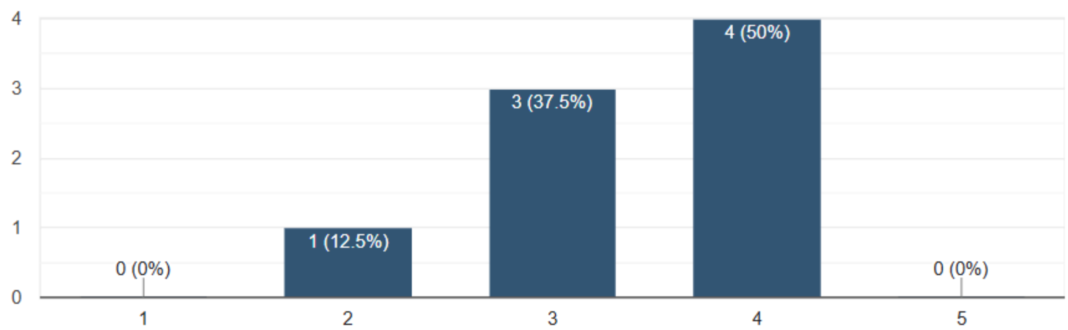
8 responses



How much do you trust AI in making security decisions?

Copy chart

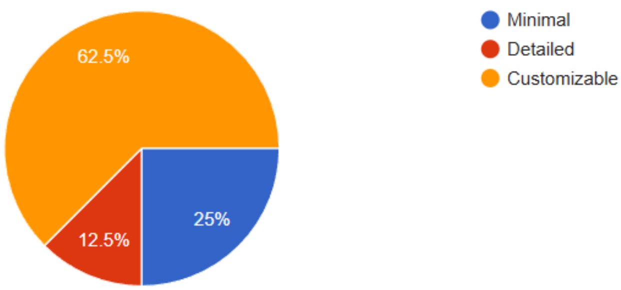
8 responses



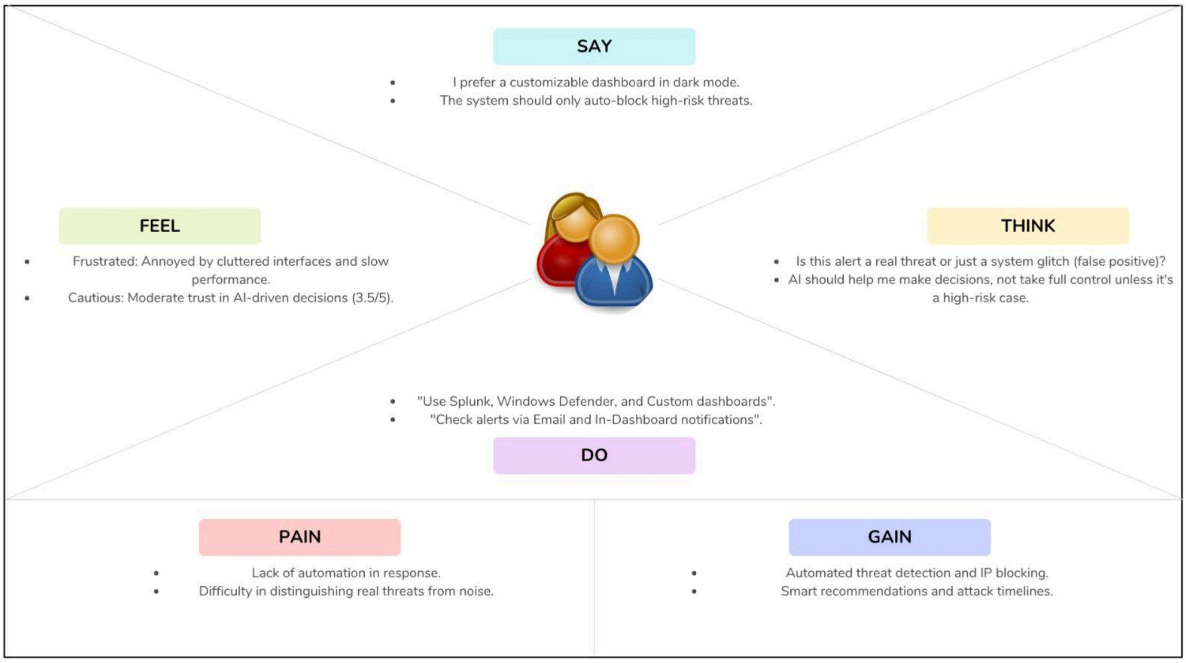
What type of dashboard layout do you prefer?

Copy chart

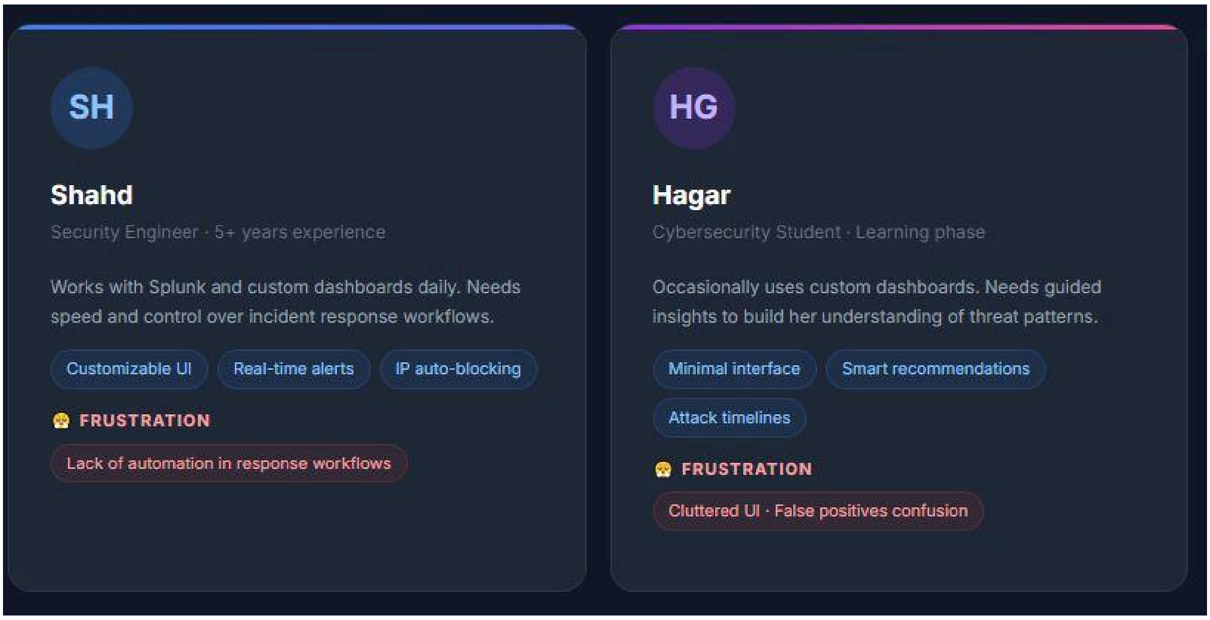
8 responses



Empathy map:



personas :



b. Stage II: Define a problem

Pain Points (Categorized)

⚡ Performance

• Slow log analysis

• Delayed alert notifications

• High manual workload

🎨 Usability

• Cluttered interfaces

• Lack of customization

• No dark mode support

🔒 Security

• No automated response

• No role separation

• False positive overload

5Ws & H Analysis

Dimension	Analysis
Who	SOC Analysts, Security Engineers, and Cybersecurity students managing network security.
What	Overwhelming volumes of alerts causing fatigue, with inability to separate real threats from false positives.
Where	In Security Operations Centers (SOCs) and enterprise IT environments.
When	During continuous 24/7 security monitoring shifts and incident response cycles.
Why	Existing tools lack intelligent automation, forcing manual review of every alert.
How	By building an AI-powered dashboard with automated threat detection, risk scoring, and RBAC.

Problem Statements

Problem Statement 1

Security analysts who monitor network activity **need a way to automatically differentiate real threats from false positives** because current tools generate excessive alerts that overwhelm teams and slow incident response.

Problem Statement 2

Junior cybersecurity students **need a simplified, guided dashboard interface** because existing enterprise tools are too complex and cluttered for learning and understanding attack patterns effectively.

Problem Statement 3

SOC teams **need automated role-based access and response workflows** because traditional dashboards lack separation between Admin and Analyst duties, creating security vulnerabilities at the access layer.

c. Stage III: Ideate a solution

Goal Statements

Goal Statement 1

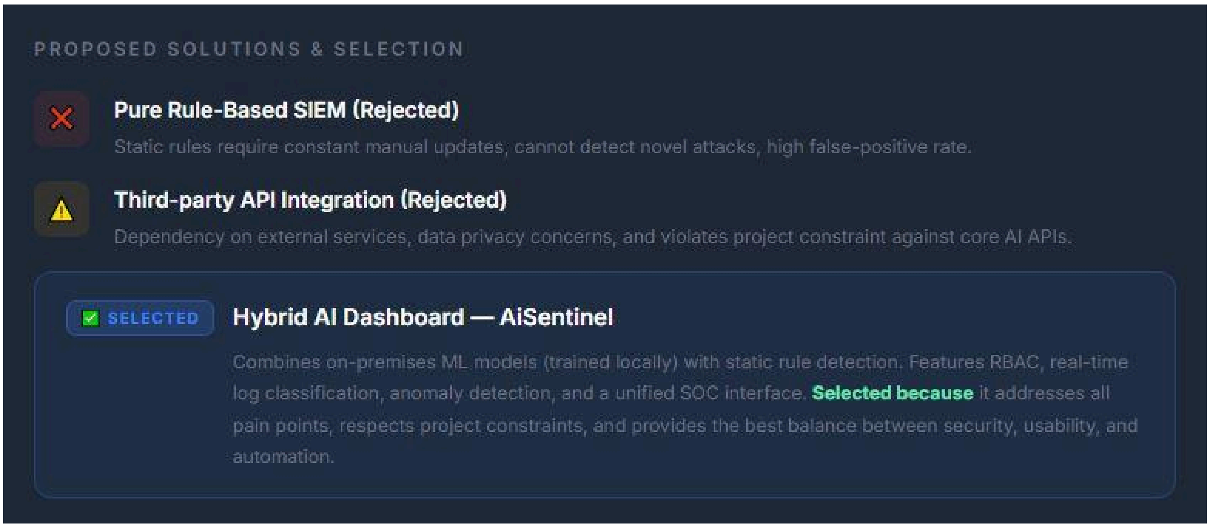
Our **AI Detection Engine** will let **security analysts automatically classify network logs as Normal or Attack in real time**, which will affect alert response time by **reducing manual analysis by 70%**. We will measure effectiveness by tracking average incident response time.

Goal Statement 2

Our **RBAC System** will let **SOC administrators control access permissions separately for Admins and Analysts**, which will affect system security by **eliminating unauthorized access incidents**. We will measure this by tracking role-based login attempts and access violations.

Goal Statement 3

Our **Unified SOC Dashboard** will let **cybersecurity students and professionals view, filter, and act on security data from a single interface**, which will affect user productivity by **replacing 3+ fragmented tools with one platform**. We will measure this by user satisfaction ratings.



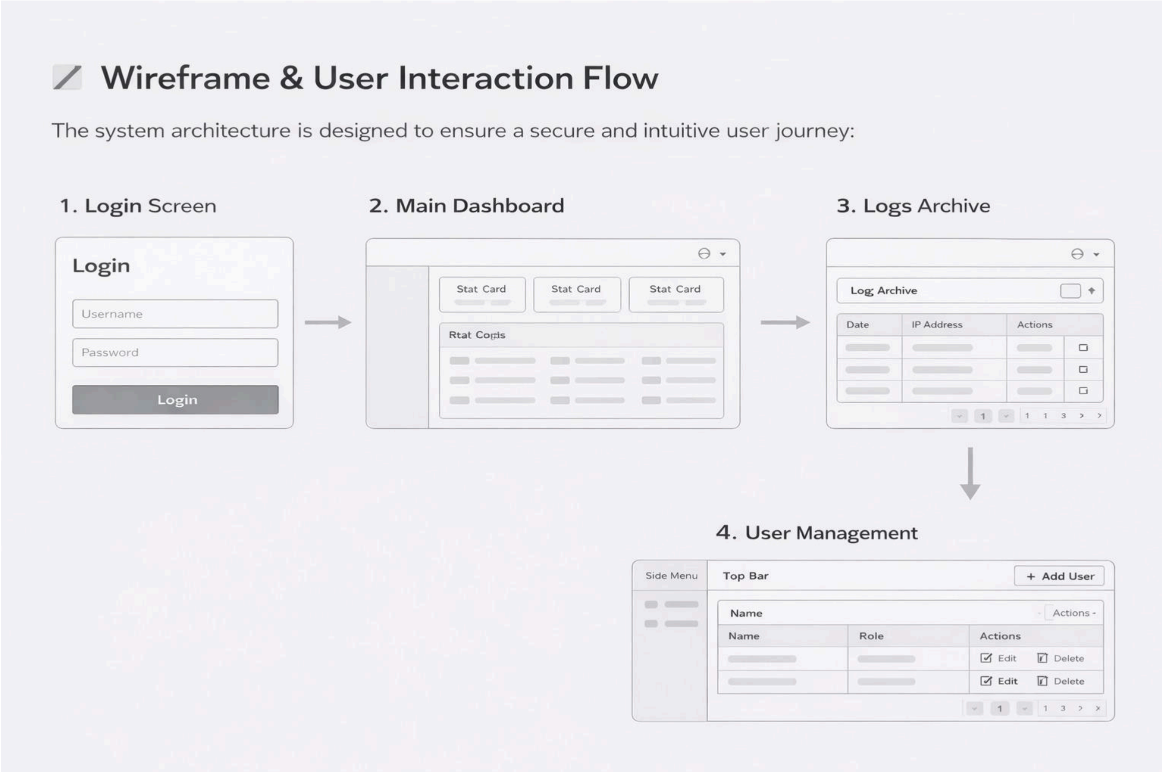
Selected Solution & Features

Selected Solution: AiSentinel — a unified AI-powered SOC dashboard with the following features:

Feature	Description
Real-time Log Analysis	Instant ML-based classification of network logs into Normal / Attack
Anomaly Detection	Detects unusual patterns beyond known attack signatures
Static Rules Engine	Configurable rule-based detection for known threat patterns
RBAC	Role-Based Access Control: Admin vs. Analyst permissions
Logs Archive	Searchable, paginated historical log viewer with filters
Dark/Light Mode	Full theme support via Tailwind CSS global toggle
User Management	Admin can add, edit, delete analyst accounts
Notifications	In-dashboard and configurable alert notifications

d. Stage IV

Wireframe Overview

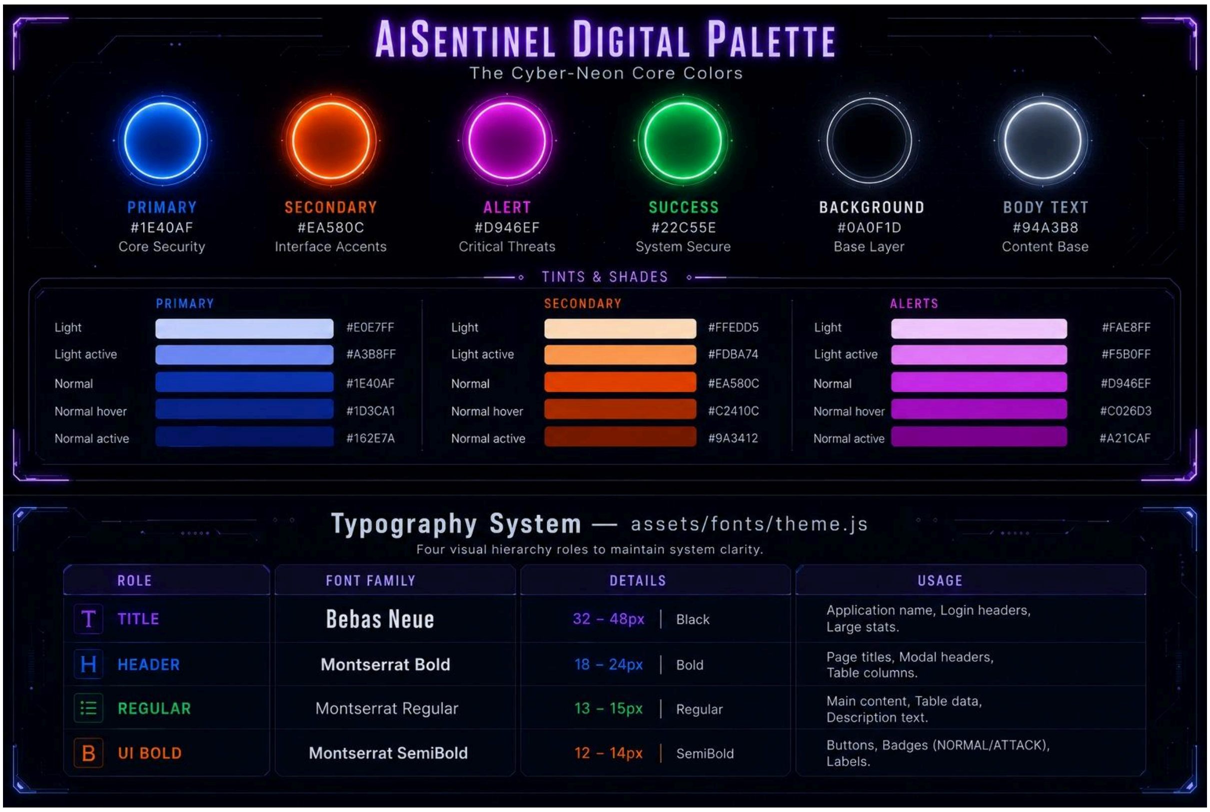


Site map that shows navigation between screens:

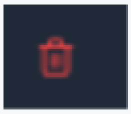
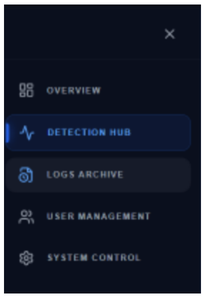
```
User Entry Point
│
└──  Authentication (LoginPage.js)
     │
     ├── Failed: [Invalid Credentials] ➔ Stay at LoginPage
     │
     └── Success: [Generate JWT Token]
         │
         └──  Initial Access Check (First Login?)
              │
              ├──  Yes: [is_first_login == True]
              │    └──  Redirect to: 🖥️ ChangePasswordPage.js (Force Update)
              │
              └── No: [Proceed to Application]
                  │
                  └── Role-Based View (RBAC Logic)
                      │
                      ├── Role: SUPER_ADMIN (Full Privilege)
                      │   ├── Access: 🖥️ Dashboard.js (Global Stats)
                      │   ├── Access: 🖥️ DetectionHub.js (Live Analysis)
                      │   ├── Access: 🖥️ LogsHistory.js (Auditing)
                      │   └──Access: 🖥️ UserManagement.js (CRUD Operations)
                      │
                      └──  Role: ANALYST (Security Monitoring)
                          ├── Access: 🖥️ Dashboard.js (System Health)
                          ├── Access: 🖥️ DetectionHub.js (Threat Hunting)
                          ├── Access: 🖥️ LogsHistory.js (Investigation)
                          └──  Denied: 🖥️ UserManagement.js (Hidden/Blocked
```

Section 02: Front-End

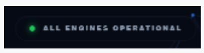
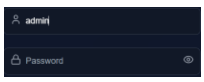
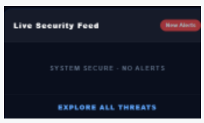
design colors (primary, secondary, and neutral) and fonts:



Custom Buttons Table

Component Name	Style (Tailwind/Visual)	Where is used in the app	Component Design
PrimaryActionButton	Neon blue background, bg-blue-600, bold uppercase text, white font, hover:shadow-[0_0_15px_#2563eb].	**Login Page, User Management (Save/Add User).**	
DangerZoneButton	Subtle red tint, bg-red-500/10, red border border-red-500/20, text-red-500, hover glow red.	**Detection Hub (Block IP), User Management (Delete User).**	
GhostNavButton	Transparent, text-gray-400, hover:bg-white/5, active state with text-blue-400 and blue left border.	**Sidebar (Navigation Items).**	
IconButton	Square rounded, p-2, backdrop-blur-xl, border-white/5, neon icon color transition.	**Header (Notifications, Profile toggle, Sidebar toggle).**	

Other Custom Component

Component Name	Style (Tailwind/Visual)	Where is used in the app	Component Design
CyberStatCard	Dark glassmorphism, bg-[#0a0f1d]/50, neon border, contains icon + label + value.	**Dashboard Overview (Top stats grid).**	
SecurityBadge	Small, px-4 py-1.5, rounded-lg, tracking-widest, colors: Green (Normal) / Red (Attack).	**Logs Archive (Status column), Detection Hub.**	
GlassInput	Semi-transparent black, bg-black/20, border-white/10, blue focus ring, placeholder:text-gray-600.	**Login Page, Archive (Search bar), User Modals.**	
TooltipNeon	Floating, bg-[#0a0f1d], border blue-500/20, slide animation, includes pointer arrow.	**Sidebar (When collapsed - shows page name).**	

AISENTINEL

MODULAR PROTECTION SYSTEM

C:\AiSentinel

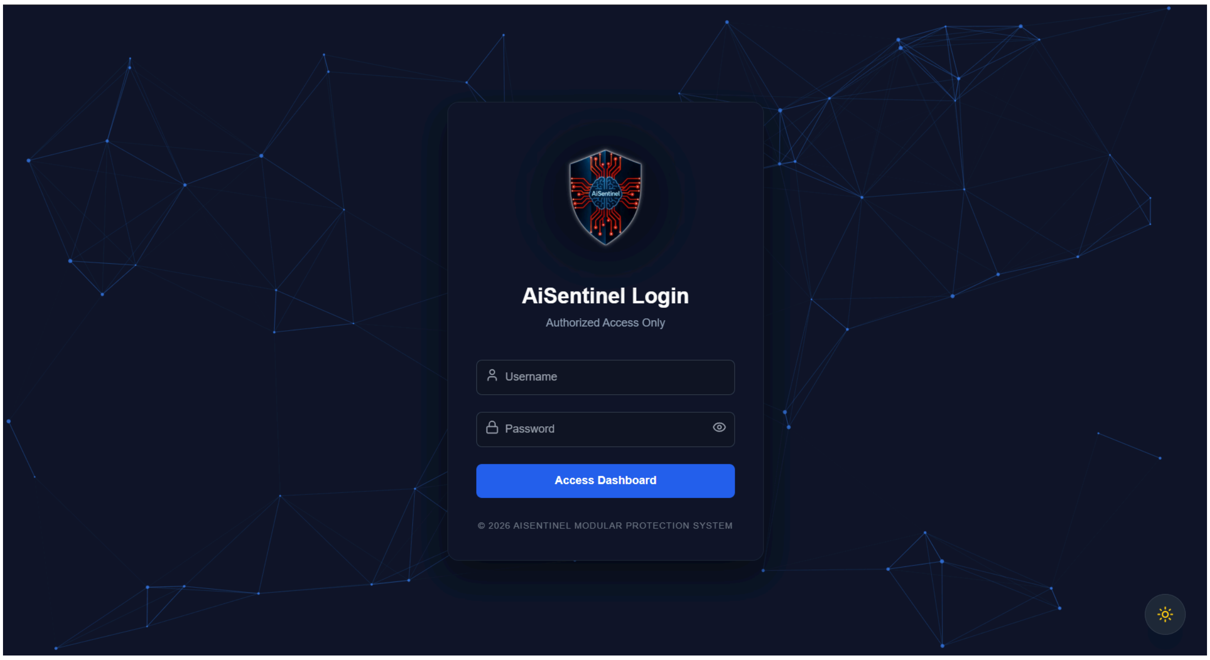
STRUCTURE	DESCRIPTION	TECHNOLOGIES
activeResponse └ active_responder.py	Active response engine for real-time threat mitigation.	Python
anomaly ├ artifacts ├ inference_pipeline.py └ trainer.py	Anomaly detection module with training and inference pipelines.	Python Scikit-learn
api ├ main_api.py ├ auth.py └ create_db.py	FastAPI backend for authentication, routes, and database setup.	FastAPI SQLAlchemy
core ├ static_engine.py ├ redis_client.py └ config.py	Core utilities: static rules engine, Redis client, and global config.	Python Redis

dashboard
├ node_modules
├ public
└ src
  ├ assets
  ├ components
  ├ hooks
  ├ pages
  │ ├ Dashboard.js
  │ ├ DetectionHub.js
  │ ├ LogsHistory.js
  │ ├ UserManagement.js
  │ ├ LoginPage.js
  │ └ ChangePasswordPage.js
  ├ utils
  ├ App.js
  └ index.js
	React frontend dashboard for monitoring, alerts, users, and system visualization.	React
		JavaScript
		Tailwind CSS
		Chart.js

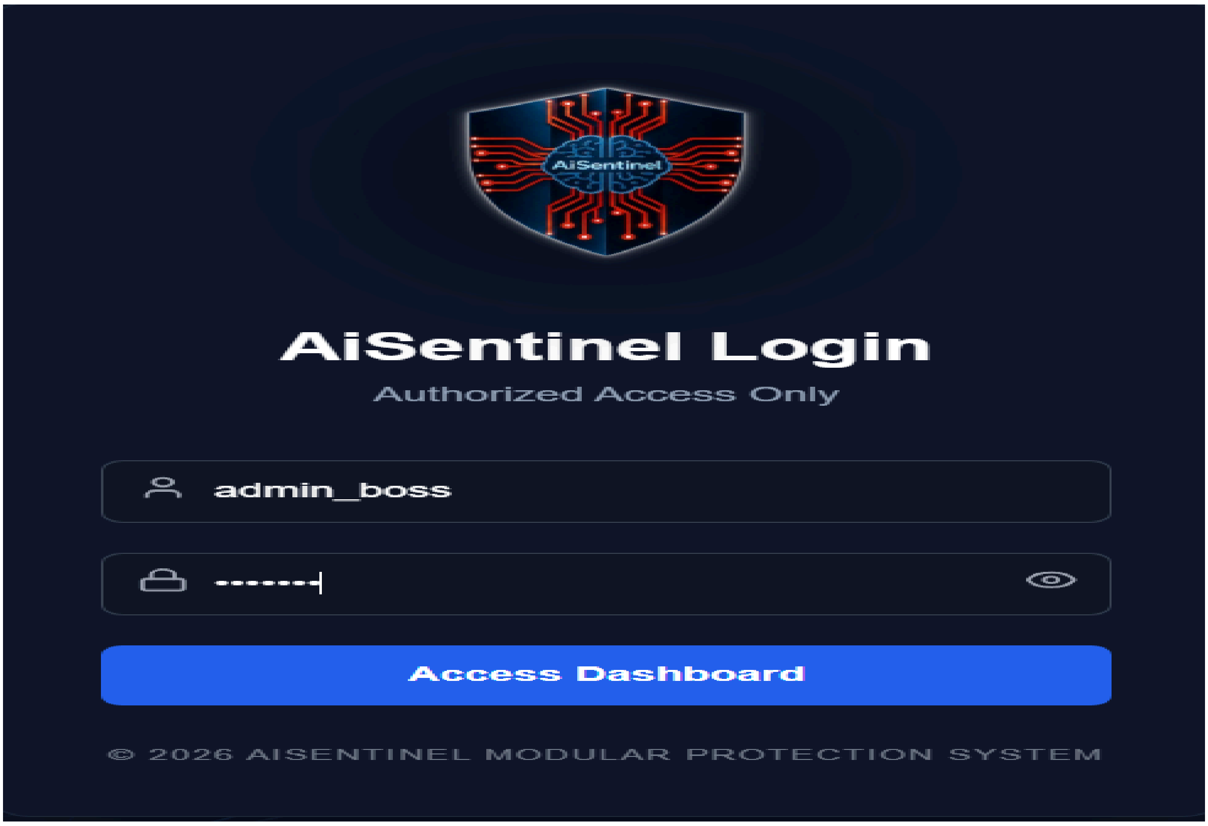
data ├ data.csv ├ preprocessing.py └ data_loader.py	Data handling: raw data, preprocessing, and loading.	Python Pandas
security └ security_pipeline.py	Security analysis pipeline for log and event analysis.	Python
testing ├ dos_test.py ├ brute_force_test.py ├ final_attack_test.py └ test_csv.py	Attack simulations and system testing scripts.	Python Requests
threat ├ artifacts └ inference_pipeline_td.py	Threat detection module with trained models.	Python Scikit-learn
tools ├ filebeat ├ logstash └ elasticsearch	ELK stack tools for log collection and analysis.	ELK Stack (Elastic)
aisentinal_raw_data.jsonl	Raw collected dataset (JSONL format).	JSONL
security_detections.csv	Exported security detections.	CSV
run.py	Main runner script to start the system.	Python
users.db	SQLite database for users and auth.	SQLite

Screenshots for UI design screens & some examples of your app responsiveness.

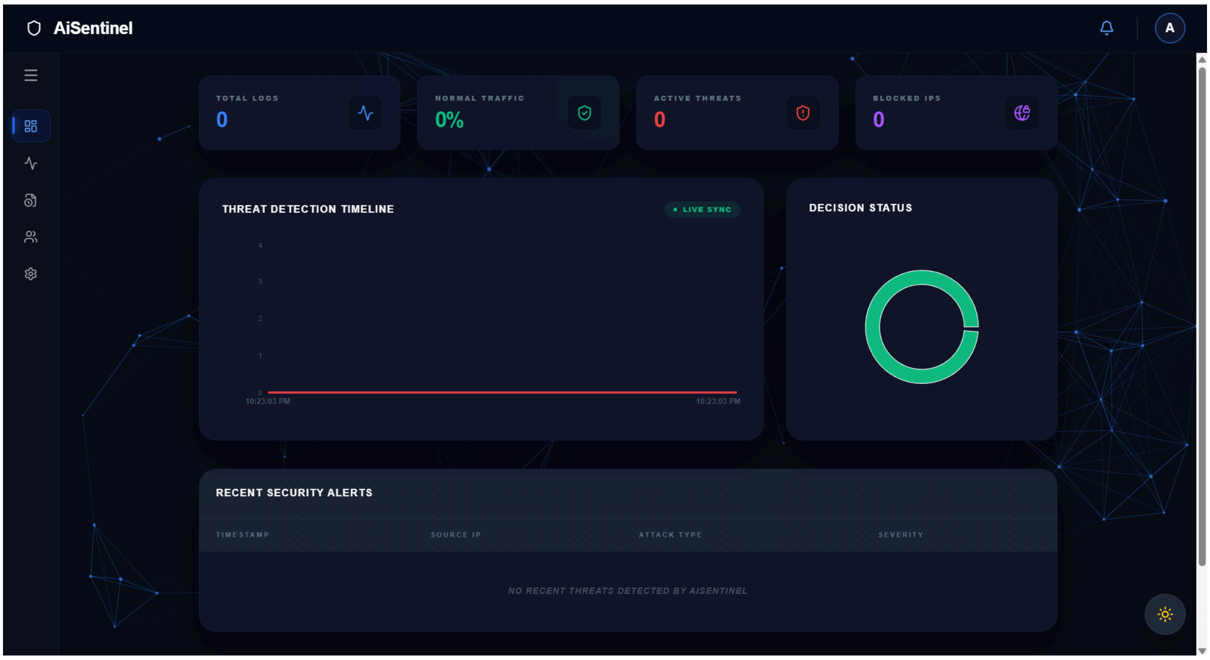
1. Login Interface (Full) : A dark-themed, responsive authentication gateway featuring high-fidelity glassmorphism and secure credential validation.



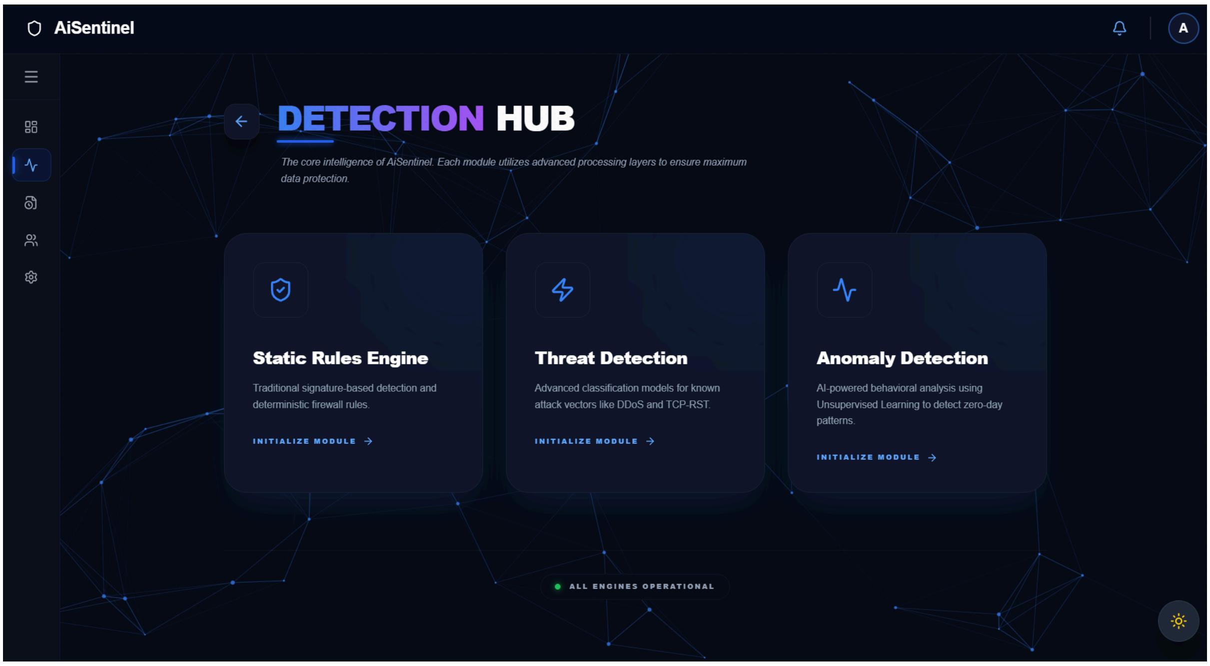
2. Authentication Form : Focused view of the secure login module with integrated identity verification and encrypted input fields.



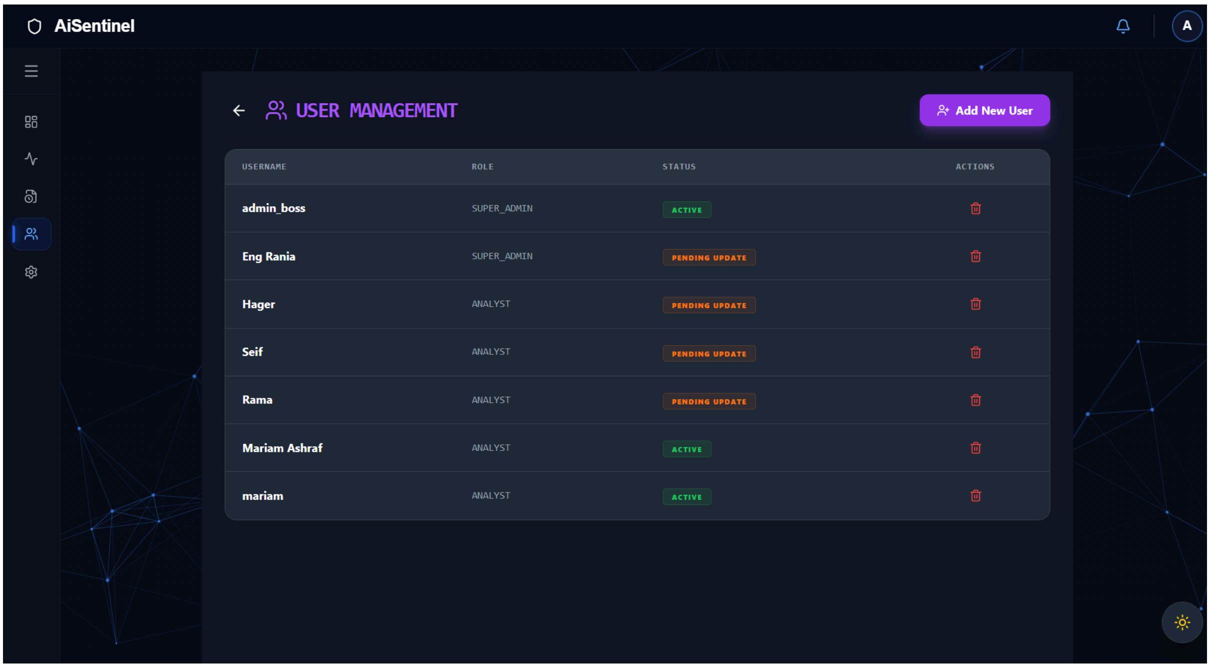
3. Global Dashboard : Centralized monitoring hub providing real-time visual telemetry, threat detection timelines, and key performance security metrics.



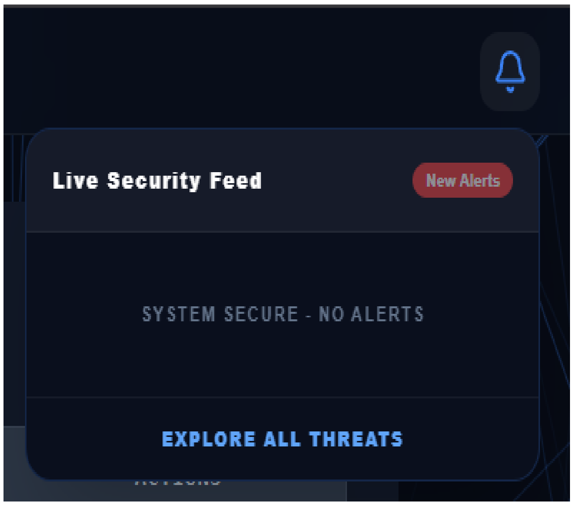
4. Detection Hub : The core intelligence interface displaying modular AI engines (Static, Threat, and Anomaly) for multi-layered traffic analysis.



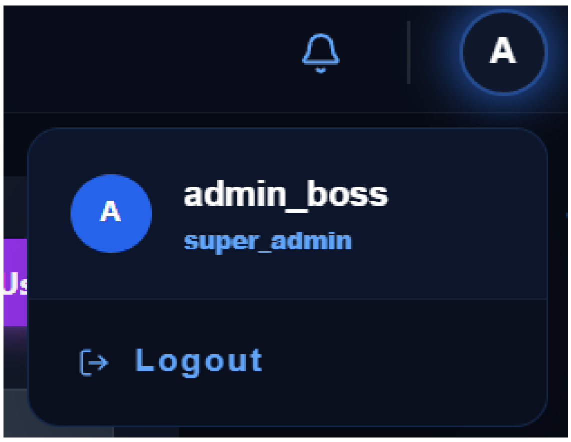
5. User Management Panel : Administrative console for Role-Based Access Control (RBAC), displaying user status, roles, and account lifecycle management.



6. Live Security Feed : Dynamic dropdown component utilizing WebSockets to push real-time security alerts and system status notifications to the analyst.



7.User Profile & Identity : Integrated identity module displaying the active session user, role assignment (e.g., Super Admin), and secure logout functionality.



Section 03: Backend

Tables (2)		
security_alerts		CREATE TABLE security_alerts
timestamp	TEXT	"timestamp" TEXT
source	TEXT	"source" TEXT
ip	TEXT	"ip" TEXT
reason	TEXT	"reason" TEXT
severity	TEXT	"severity" TEXT
raw_log	TEXT	"raw_log" TEXT
users		CREATE TABLE users (usernan
username	TEXT	"username" TEXT
hashed_password	TEXT	"hashed_password" TEXT
role	TEXT	"role" TEXT
must_change_password	INTEGER	"must_change_password" INT

	timestamp	source	ip	reason	severity	raw_log
	Filter	Filter	Filter	Filter	Filter	Filter
1	2026-04-27 17:38:11	Static Engine	127.0.0.1	MULTIPART_ATTACK: Multipart header …	High	{"source_ip": "127.0.0.1", …
2	2026-04-27 18:15:56	Static Engine	127.0.0.1	MULTIPART_ATTACK: Multipart header …	High	{"source_ip": "127.0.0.1", …
3	2026-04-27 18:19:28	Static Engine	127.0.0.1	MULTIPART_ATTACK: Multipart header …	High	{"source_ip": "127.0.0.1", …
4	2026-04-27 20:29:57	Static Engine	127.0.0.1	MULTIPART_ATTACK: Multipart header …	High	{"source_ip": "127.0.0.1", …
5	2026-04-27 20:31:56	Static Engine	127.0.0.1	MULTIPART_ATTACK: Multipart header …	High	{"source_ip": "127.0.0.1", …
6	2026-04-27 20:42:21	Static Engine	127.0.0.1	MULTIPART_ATTACK: Multipart header …	High	{"source_ip": "127.0.0.1", …
7	2026-04-27 20:51:30	Static Engine	127.0.0.1	MULTIPART_ATTACK: Multipart header …	High	{"source_ip": "127.0.0.1", …
8	2026-04-27 20:55:28	Static Engine	127.0.0.1	MULTIPART_ATTACK: Multipart header …	High	{"source_ip": "127.0.0.1", …
9	2026-04-27 21:01:31	Static Engine	127.0.0.1	LFI_RFI: Possible Remote File …	Critical	{"source_ip": "127.0.0.1", …
10	2026-04-27 21:22:06	Static Engine	127.0.0.1	LFI_RFI: Possible Remote File …	Critical	{"source_ip": "127.0.0.1", …
11	2026-04-27 21:25:32	Static Engine	127.0.0.1	LFI_RFI: Possible Remote File …	Critical	{"source_ip": "127.0.0.1", …
12	2026-04-27 21:29:05	Static Engine	127.0.0.1	LFI_RFI: Possible Remote File …	Critical	{"source_ip": "127.0.0.1", …
13	2026-04-27 21:35:54	Static Engine	127.0.0.1	LFI_RFI: Possible Remote File …	Critical	{"source_ip": "127.0.0.1", …
14	2026-04-27 21:36:25	Static Engine	127.0.0.1	LFI_RFI: Possible Remote File …	Critical	{"source_ip": "127.0.0.1", …
15	2026-04-27 21:38:17	Static Engine	127.0.0.1	LFI_RFI: Possible Remote File …	Critical	{"source_ip": "127.0.0.1", …
16	2026-04-27 21:59:55	Static Engine	127.0.0.1	PROTOCOL_VIOLATION: Range: Invalid …	Medium	{"source_ip": "127.0.0.1", …
17	2026-04-27 22:05:12	Static Engine	127.0.0.1	PROTOCOL_VIOLATION: Content-Type …	Critical	{"source_ip": "127.0.0.1", …
18	2026-04-27 22:06:56	Static Engine	127.0.0.1	PROTOCOL_VIOLATION: Range: Invalid …	Medium	{"source_ip": "127.0.0.1", …
19	2026-04-27 22:08:29	Static Engine	127.0.0.1	PROTOCOL_VIOLATION: Range: Invalid …	Medium	{"source_ip": "127.0.0.1", …
20	2026-04-27 22:11:56	Static Engine	127.0.0.1	LFI_RFI: Possible Remote File …	Critical	{"source_ip": "127.0.0.1", …
21	2026-04-27 22:18:55	Static Engine	127.0.0.1	LFI_RFI: Possible Remote File …	Critical	{"source_ip": "127.0.0.1", …
22	2026-04-29 13:09:31	Static Engine	127.0.0.1	SQL_INJECTION	High	{"source_ip": "127.0.0.1", …
23	2026-04-29 13:16:54	Static Engine	127.0.0.1	SQL_INJECTION	High	{"source_ip": "127.0.0.1", …
24	2026-04-29 19:36:58	Static Engine	127.0.0.1	SQL_INJECTION	Medium	{"source_ip": "127.0.0.1", …
25	2026-04-29 19:37:17	Static Engine	127.0.0.1	XSS	Medium	{"source_ip": "127.0.0.1", …
26	2026-04-29 19:45:20	Static Engine	127.0.0.1	XSS	High	{"source_ip": "127.0.0.1", …
27	2026-04-29 19:47:05	Static Engine	127.0.0.1	SQL_INJECTION	High	{"source_ip": "127.0.0.1", …

	username	hashed_password	role	must_change_password
	Filter	Filter	Filter	Filter
1	admin_boss	$argon2id$v=19$m=65536,t=3,p=4$2vZyJ…	super_admin	0
2	Eng Rania	$argon2id$v=19$m=65536,t=3,p=4$wT2xv…	super_admin	1
3	Hager	$argon2id$v=19$m=65536,t=3,p=4$RdJ8x…	analyst	1
4	Seif	$argon2id$v=19$m=65536,t=3,p=4$Z7/…	analyst	1
5	Rama	$argon2id$v=19$m=65536,t=3,p=4$H2iKl…	analyst	1
6	Mariam Ashraf	$argon2id$v=19$m=65536,t=3,p=4$uOaKc…	analyst	0
7	mariam	$argon2id$v=19$m=65536,t=3,p=4$h/…	analyst	0

Backend part including the dynamic content:

Operation	Backend Logic	Automated Action
Real-time Threat Detection	Analyzes network traffic via AI engines and matches patterns with trained models (Inference).	Broadcasts instant alerts via WebSockets and updates the Dashboard UI in real-time.
Active Response Execution	Evaluates threat severity scores against pre-defined security thresholds.	Automatically triggers IP blocking and isolation protocols via the activeResponse module.
Authentication Guard	Monitors login attempts and validates the is_first_login attribute for each user.	Enforces mandatory password resets for new users and implements IP throttling for failed attempts.
Dynamic Content Management	Validates user privileges (RBAC) stored in the encrypted users.db database.	Dynamically injects system modules and accessible routes based on the authenticated user's role.
Background Data Pipeline	Cleans and transforms raw network logs into numeric feature matrices (Label Encoding).	Automatically prepares and feeds formatted data into AI models for high-speed prediction.

Table that specifies all the technologies used for frontend and backend:

Technology	Role	Module	Layer
Frontend			
React	UI component framework	dashboard/src	Frontend
JavaScript	Application logic & interactivity	dashboard/src	Frontend
Tailwind CSS	Utility-first styling framework	dashboard/src	Frontend
Chart.js	Data visualization & charting	dashboard/src	Frontend
Backend			
Python	Core language across all modules	core, api, security, activeResponse, testing	Backend
FastAPI	REST API framework & routing	api/	Backend
Redis	In-memory caching & pub/sub	core/redis_client.py	Backend
SQLite	User & auth persistence	users.db	Backend
Requests	HTTP client for attack simulation	testing/	Backend
Data & ML			
Scikit-learn	ML model training & inference	anomaly/, threat/	Data / ML
Pandas	Data loading & preprocessing	data/	Data / ML
JSONL / CSV	Raw dataset & detection export formats	aisentinel_raw_data.jsonl, security_detections.csv	Data / ML
Infrastructure			
Filebeat	Log shipping agent	tools/filebeat	Infrastructure
Logstash	Log processing pipeline	tools/logstash	Infrastructure